UNIT 4

LAYERS OF EARTH

Learning objectives:

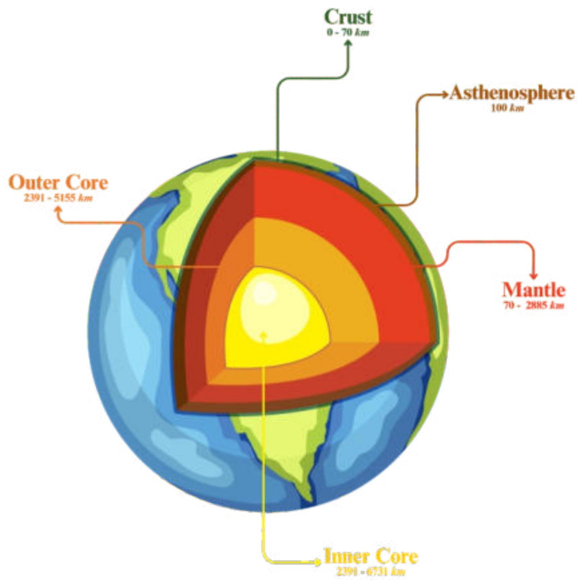
- Identify and describe the four main layers of the Earth.
- Differentiate between continental crust and oceanic crust in terms of composition and thickness.
- Explain the structure and properties of the mantle, including its division into the upper and lower mantle, and how its movement influences tectonic plate shifts.
- Describe the composition and states of the core (outer core as liquid and inner core as solid) and their roles in generating Earth's magnetic field.
- Understand the interactions between Earth's layers and their role in geological events like earthquakes, volcanic eruptions, and mountain formation.

EARTH'S LAYERS

Earth is a unique planet that supports life, thanks to its structure and composition. It is divided into four main layers: the crust, mantle, outer core, and inner core. These layers are distinct in their characteristics, composition, and function. From the solid crust where we live to the molten metal in the core, each layer plays a crucial role in shaping Earth's surface and processes like earthquakes, volcanic eruptions, and mountain formation.

THE CRUST

The crust is the outermost layer of the Earth. It is where all life exists, including humans, animals, and plants. The crust is made of solid rock and soil. It is the thinnest layer, with an average thickness of about 30 kilometres on land (continental crust) and about 5-10 kilometres under the oceans (oceanic crust).



- **Continental crust:** Made mostly of granite, it forms continents.
- **Oceanic crust:** Made mostly of basalt, it forms the ocean floor.

THE MANTLE

Beneath the crust lies the mantle, the thickest layer of the Earth, extending about 2,900 kilometres deep. It is made of semi-solid, hot rocks that move slowly. This movement is responsible for the shifting of tectonic plates, which can cause earthquakes and volcanic eruptions. The mantle has two main parts:

- **Upper Mantle:** The part closest to the crust, semi-solid and cooler.
- **Lower Mantle:** Hotter and more fluid than the upper mantle.

THE CORE

The core is divided into two parts:

- **Outer Core:** Made of liquid iron and nickel, it is about 2,200 kilometres thick. The movement of this liquid creates Earth's magnetic field.
- **Inner Core:** A solid ball of iron and nickel, about 1,200 kilometres thick. The immense pressure keeps it solid despite the high temperatures, estimated to be as hot as the surface of the Sun (5,000-6,000°C).

HOW THESE LAYERS INTERACT

The interaction between these layers causes geological events:

- **Tectonic Plate Movements**: The mantle's slow movement pushes plates in the crust, causing earthquakes and the formation of mountains.
- **Volcanoes**: Magma from the mantle rises through cracks in the crust to form volcanoes.

Magnetic Field: The movement of the liquid outer core generates Earth's magnetic field, protecting us from harmful solar radiation.

Fill in the blanks.

1. The outermost layer of the Earth is called the _______________ .

2. The inner core is _______________ because of the high pressure.

3. Earthquakes are caused by movements in the _______________.

4. _______________ is made mostly of granite. It forms continent.

5. The outer core is responsible for Earth's _______________.

6. The mantle accounts for _______________ of Earth's volume.

7. The inner core is as hot as the surface of the _______________.

8. The Earth's crust is made of solid _______________ and _______________.

Choose the correct answer.

1. Which is the thinnest layer of the Earth?

- Mantle
- Crust
- Outer Core
- Inner Core

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2. What is the mantle mostly made of?

- Iron
- Semi-solid rock
- Water
- Soil

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3. The Earth's core is divided into:

- Mantle and crust
- Upper and lower mantle
- Outer and inner core
- Crust and mantle

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4. What generates Earth's magnetic field?

- Inner core
- Mantle
- Outer core
- Crust

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5. Which crust is thicker?

- Oceanic crust
- Continental crust
- Both are the same thickness
- Crust under mountains

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6. The hottest layer of the Earth is:

- Iron
- Mantle
- Outer core
- Inner core

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7. The crust is divided into:

- Two types
- Three types
- Four types
- One type

Fun Fact

The mantle accounts for 84% of Earth's total volume!

Answer the following questions.

Q1: Name the four layers of the Earth.

Q2: Describe the two characteristics of “Crust” and “Mantle”.

Q3: Why is the inner core solid despite being the hottest layer?

Q4: How do movements in the mantle affect the Earth's surface?

Q5: What is the difference between the oceanic crust and the continental crust?

Critical Thinking

What would happen if the Earth's mantle stopped moving?

How would it affect life on the surface?

Encourage students to discuss the effects on tectonic plate movement, earthquakes, volcanoes, and the formation of mountains.

Make Your Own Earth Layers Model

ACTIVITY

Visualize the Earth's layers and their arrangement.

- **Materials Needed**: Playdough or modeling clay (different colours), a knife, and labels.
- **Instructions**:

1. Use different colours of clay to represent the crust, mantle, outer core, and inner core.
2. Stack them to create a layered Earth.
3. Cut the model in half to show a cross-section.
4. Label each layer.

- **Outcome**: A tangible 3D model of Earth's layers.